

Automatidata – NYC Taxi Data Inspection

Foundations of Data Science

Performed an initial inspection of the New York City Taxi & Limousine Commission dataset using Python and pandas. The dataset was imported, reviewed for data types, summary statistics, and potential data quality issues to prepare it for exploratory data analysis (EDA).

Key Insights

- Imported the taxi dataset into a pandas Data Frame.
- Reviewed column data types and non-null values.
- Generated descriptive statistics using `describe()`.
- Investigated trip distance and total fare values.
- Compared payment methods and vendor information.
- Created filtered datasets for additional analysis.

Details

```
df.info()

[20]

... <class 'pandas.core.frame.DataFrame'>
RangeIndex: 22699 entries, 0 to 22698
Data columns (total 18 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Unnamed: 0                            22699 non-null  int64
1   VendorID                              22699 non-null  int64
2   tpep_pickup_datetime                  22699 non-null  object
3   tpep_dropoff_datetime                  22699 non-null  object
4   passenger_count                        22699 non-null  int64
5   trip_distance                          22699 non-null  float64
6   RatecodeID                            22699 non-null  int64
7   store_and_fwd_flag                    22699 non-null  object
8   PULocationID                          22699 non-null  int64
9   DOLocationID                          22699 non-null  int64
10  payment_type                           22699 non-null  int64
11  fare_amount                            22699 non-null  float64
12  extra                                 22699 non-null  float64
13  mta_tax                               22699 non-null  float64
14  tip_amount                            22699 non-null  float64
15  tolls_amount                           22699 non-null  float64
16  improvement_surcharge                  22699 non-null  float64
17  total_amount                           22699 non-null  float64
dtypes: float64(8), int64(7), object(3)
memory usage: 3.1+ MB
```

Next Steps

Check for missing values and duplicate records. Perform exploratory data analysis. Identify relationships between trip variables. Prepare the data for predictive modeling.